Evaluation Metrics for Natural Language Processing

ANLP Assignment 2

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Contents

1 Reference-Based Evaluation Metrics **2**

1.1 ROUGE (Recall-Oriented Understudy for Gisting Evaluation) 2

1.2 BLEU (Bilingual Evaluation Understudy) 2

1.3 BERTScore 2

2 Reference-Free Evaluation Approaches **3**

3 Bonus: A Proposed Composite Evaluation Metric **4**

1 Reference-Based Evaluation Metrics

1.1 ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

ROUGE serves as a standard metric for evaluating automatic summarization systems by measuring recall—essentially how successful an AI-generated summary was in capturing essential information from human-written reference summaries.

Primary Variants

- **ROUGE-N:** measures n-gram overlap between generated and reference summaries. ROUGE-1 analyzes unigram overlap, examining individual word matches, while ROUGE-2 evaluates bigram overlap, considering pairs of consecutive words.
- **ROUGE-L:** employs the Longest Common Subsequence (LCS) algorithm to identify the longest sequence of words appearing in both summaries in the same order (subsequence, regardless of intervening gaps). This approach better captures sentence-level structural similarities.

Strengths and Limitations

ROUGE has achieved widespread adoption due to its computational efficiency and reasonable correlation with human judgments across large datasets. However, the metric exhibits significant limitations. Most critically, ROUGE relies exclusively on lexical overlap without semantic comprehension. Consequently, summaries employing synonyms receive lower scores despite maintaining semantic equivalence. For instance, ”large vehicle” and ”big car” would be evaluated as entirely distinct.

Additionally, ROUGE may assign favorable scores to incoherent outputs provided they contain appropriate keywords, regardless of overall readability. The metric's reliability also depends heavily on reference summary quality, introducing potential evaluative bias.

1.2 BLEU (Bilingual Evaluation Understudy)

BLEU emphasizes precision over recall, assessing the proportion of machine-generated phrases that appear in reference translations.

Methodology

BLEU calculates modified n-gram precision for n-grams ranging from one to four words in length. The modification includes a clipping mechanism preventing system manipulation through excessive repetition of correct terms. A brevity penalty discourages artificially short outputs that might achieve high precision while omitting essential information.

Strengths and Limitations

BLEU's widespread adoption stems from its speed, language independence, and reasonable correlation with human judgment at the corpus level. Nevertheless, it shares ROUGE's fundamental weakness: absence of semantic understanding. Translations using alternative yet valid word choices receive penalties despite maintaining accuracy.

Furthermore, BLEU scores for individual sentences lack reliability; meaningful assessment requires evaluation across hundreds or thousands of sentences. The metric does not directly measure grammaticality, potentially assigning acceptable scores to grammatically flawed outputs containing correct phrases.

1.3 BERTScore

BERTScore represents a contemporary approach utilizing contextual embeddings from transformer models like BERT to address semantic limitations inherent in traditional metrics. Rather than treating words as discrete tokens, BERTScore compares contextual meanings.

Methodology

The evaluation process generates contextual embeddings for each token in both candidate and reference texts using BERT. Cosine similarity calculations between all token pairs enable greedy matching to determine optimal alignment. These similarity scores are then aggregated to produce precision, recall, and F1 measures at the semantic level.

Strengths and Limitations

BERTScore constitutes a substantial advancement by recognizing semantic relationships, such as understanding that "automobile" and "car" convey similar meanings. This semantic awareness produces significantly stronger correlation with human quality judgments.

However, BERTScore presents certain drawbacks. The computational requirements increase substantially due to inference processing through large transformer models. More problematically, BERTScore may overlook critical factual distinctions. For example, "The meeting is on Tuesday" and "The meeting is on Thursday" would receive high similarity scores despite conveying different information. The metric also struggles with negation: "The defendant is guilty" and "The defendant is not guilty" appear similar structurally despite expressing opposite meanings.

2 Reference-Free Evaluation Approaches

Reference-free evaluation methods assess generated text without comparison to gold-standard human references. Instead, these metrics evaluate intrinsic textual properties such as factual consistency, coherence, or fluency. This approach proves particularly valuable for tasks like open-ended dialogue generation, where multiple valid responses exist.

Example: Factual Consistency Metrics

SummaC (Summary Consistency) exemplifies this approach by evaluating whether summaries maintain factual consistency with source documents. Rather than comparing against reference summaries, SummaC directly verifies generated summary claims against original source text. The system decomposes summaries into discrete factual claims and determines whether source documents entail each claim.

Comparative Advantages

Reference-free metrics eliminate the need for expensive human-written references. Creating high-quality reference texts demands considerable time, expertise, and financial resources—reference-free approaches remove this constraint entirely.

These metrics can identify problems that reference-based evaluations miss entirely. A summary might achieve an excellent BERTScore by closely matching reference summary style and content while containing hallucinated information absent from the source document. Reference-free factual consistency metrics would detect such discrepancies.

For creative or open-ended tasks, reference-free evaluation becomes practically necessary. When generating dialogue responses or creative writing, numerous valid outputs exist, making single-reference comparison inadequate. Reference-free metrics can instead evaluate properties like coherence, appropriateness, or relevance.

Limitations

The primary concern involves reliance on additional models serving as evaluators. Factual consistency metrics typically employ natural language inference (NLI) or question-answering models to verify claims. These evaluator models possess their own limitations, biases, and failure modes, effectively substituting one challenge for another.

Reference-free metrics also tend toward a narrow focus. SummaC addresses factual consistency but provides no information regarding writing quality, conciseness, or appropriate emphasis. Comprehensive evaluation typically requires multiple reference-free metrics, increasing complexity and computational demands.

Finally, these approaches generally require more computational resources than simple n-gram overlap metrics, though typically less than BERTScore due to their specialized nature.

3 Bonus: A Proposed Composite Evaluation Metric

In the evaluation of machine-generated text, a clear dichotomy has emerged between two main approaches. The first approach uses reference-based metrics like BERTScore to measure semantic quality and similarity to a human-written ideal. The second approach uses reference-free metrics like SummaC to check for factual accuracy against an original source document. This split creates a challenge, as a model can perform well on one dimension while failing on the other.

To bridge this gap, I propose a framework for a unified score that assesses both aspects simultaneously. I call this framework the Unified Semantic and Factual Score (USFS). Its purpose is to combine a score for textual quality with a score for factual accuracy. It consists of two components:

Semantic Score (S_{qual}): This part answers the question, "Is the generated text high-quality and does it capture the right meaning?" For this, I propose using BERTScore. It compares the AI's text to a human-written example and gives a score based on how similar they are in meaning, not just words. This serves as the quality check.

Factual Score (F_{cons}): This part answers the question, "Is the information in the text correct?" For this, I propose using a reference-free tool like SummaC. It compares the AI's text directly against the original source document to check if any claims contradict the source. This serves as the fact-check.

At first, I considered a simple weighted average of these two scores. But this is not very informative as it doesn't take into account that a summary that excels at one dimension but fails at the other is fundamentally broken. A factually perfect but incomprehensible summary is useless. A beautifully written but factually wrong summary is actually dangerous, especially in domains like medical or legal text.

This insight led me to a multiplicative approach. The final USFS score uses a geometric mean with adjustable emphasis:

$$USFS = (S_{\text{qual}}^{\alpha} \times F_{\text{cons}}^{1-\alpha})$$

The multiplicative combination ensures that both dimensions must be reasonably good for the overall score to be high. If either score drops to zero, the entire USFS becomes zero—which makes intuitive sense. You can't compensate for complete factual incorrectness with beautiful writing, and vice versa.

The Flexibility Parameter (α)

The α parameter still acts as a slider (0 to 1) that lets you emphasize which dimension matters more for your task:

- $\alpha = 0.7$: Puts more weight on semantic quality (70% influence). Good for creative or stylistic tasks like story summarization.
- $\alpha = 0.3$: Puts more weight on factual consistency (70% influence). Good for news, scientific papers, or medical summaries.
- $\alpha = 0.5$: Treats both equally. Good as a general-purpose default.

But unlike a linear combination, even with a high α, a catastrophic failure on the factual side will still significantly hurt your score.

Advantages

1. **Addresses the Complementary Weakness Problem:** Existing metrics have blind spots. BERTScore can give high scores to fluent hallucinations, while SummaC doesn't care if the text is repetitive, incoherent, or awkwardly phrased. USFS requires competence on both dimensions, making it harder to "game" with a model that's good at one thing but terrible at another.

2. **More Reflective of Human Judgment:** It tries to quantify more qualitative judgments about text quality in a way that mirrors actual human evaluation. When people read a summary they consider flow, trustworthiness, and usefulness all at once. The geometric mean in USFS mimics this by producing scores that reflect overall impression rather than isolated strengths.
3. **Task Adaptability with Theoretical Grounding:** Unlike ad-hoc metric combinations, USFS has a clear interpretation: the α parameter represents the relative importance of semantic quality vs. factual consistency for your specific use case. This makes it easy to justify your choice and compare across different evaluation setups.
4. **Practical Advantages:** A single number makes it easier to rank models, especially when presenting to non-technical stakeholders. It also provides a built-in quality threshold; if you require $USFS > 0.8$, you're implicitly requiring both components to be strong.

Limitations

- **Still requires a reference summary:** BERTScore needs a human-written reference, which is expensive to obtain and can introduce variance.
- **Scale dependency:** BERTScore and SummaC might have different score distributions. It's recommended to normalize both to the same scale (e.g., 0-1) before combining them.
- **Can't detect all error types:** Neither component catches issues like:
  - Omission of critical information
  - Improper emphasis or framing
  - Subtle bias or tone problems
- **Computational cost:** Running two complex metrics is slower and more resource-intensive than one.
- **The α tuning problem:** Choosing the right α requires domain expertise and potentially its own validation against human judgments.